

INTRODUCTION TO



BUILT-IN MODULE



BY

ASHFIQUZZAMAN
SAJAL

HTTP MODULE

THE HTTP MODULE CAN CREATE AN HTTP SERVER USING `createServer()` METHOD THAT LISTENS TO SERVER PORTS AND GIVES A RESPONSE BACK TO THE CLIENT.

JS http.js

```
1 var http = require('http');
2
3 //create a server object:
4 http.createServer(function (req, res) {
5   res.write('Hello World!'); //write a response to the client
6   res.end(); //end the response
7 }).listen(8080); //the server object listens on port 8080
```



FS MODULE

THE NODE.JS FILE SYSTEM MODULE
ALLOWS YOU TO WORK WITH THE FILE
SYSTEM ON YOUR COMPUTER.

JS fs.js

```
1 var http = require('http');
2 var fs = require('fs');
3 http.createServer(function (req, res) {
4   fs.readFile('demofile1.html', function(err, data) {
5     res.writeHead(200, {'Content-Type': 'text/html'});
6     res.write(data);
7     return res.end();
8   });
9 }).listen(8080);
```



URL MODULE

THE URL MODULE SPLITS UP A WEB ADDRESS INTO READABLE PARTS.

JS url.js

```
1 var url = require('url');
2 var adr = 'http://localhost:8080/default.htm?year=2017&month=february';
3 var q = url.parse(adr, true);
4
5 console.log(q.host); //returns 'localhost:8080'
6 console.log(q.pathname); //returns '/default.htm'
7 console.log(q.search); //returns '?year=2017&month=february'
8
9 var qdata = q.query; //returns an object: { year: 2017, month: 'february' }
10 console.log(qdata.month); //returns 'february'
```



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EVENT MODULE

EVERY ACTION ON A COMPUTER IS AN EVENT. LIKE WHEN A CONNECTION IS MADE OR A FILE IS OPENED.

NODE.JS HAS A BUILT-IN MODULE, CALLED "EVENTS", WHERE YOU CAN CREATE-, FIRE-, AND LISTEN FOR- YOUR OWN EVENTS.



JS event.js

```
1  var events = require('events');
2  var EventEmitter = new events.EventEmitter();
3
4  //Create an event handler:
5  var myEventHandler = function () {
6      console.log('I hear a scream!');
7  }
8
9  //Assign the event handler to an event:
10  EventEmitter.on('scream', myEventHandler);
11
12  //Fire the 'scream' event:
13  EventEmitter.emit('scream');
```



FORMIDABLE MODULE

THERE IS A VERY GOOD MODULE FOR WORKING
WITH FILE UPLOADS, CALLED "FORMIDABLE".

```
JS formidable.js

1 var http = require('http');
2 var formidable = require('formidable');
3 var fs = require('fs');
4
5 http.createServer(function (req, res) {
6   if (req.url == '/fileupload') {
7     var form = new formidable.IncomingForm();
8     form.parse(req, function (err, fields, files) {
9       var oldpath = files.fileupload.filepath;
10      var newpath = 'C:/Users/Your Name/' + files.fileupload.originalFilename;
11      fs.rename(oldpath, newpath, function (err) {
12        if (err) throw err;
13        res.write('File uploaded and moved!');
14        res.end();
15      });
16    });
17   } else {
18     res.writeHead(200, {'Content-Type': 'text/html'});
19     res.write('<form action="fileupload" method="post" enctype="multipart/form-data">');
20     res.write('<input type="file" name="fileupload"><br>');
21     res.write('<input type="submit">');
22     res.write('</form>');
23     return res.end();
24   }
25 }).listen(8080);
```



NODEMAILER MODULE

THE NODEMAILER MODULE MAKES IT EASY TO SEND
EMAILS FROM YOUR COMPUTER.



JS nodemailer.js

```
1  var nodemailer = require('nodemailer');
2
3  var transporter = nodemailer.createTransport({
4    service: 'gmail',
5    auth: {
6      user: 'youremail@gmail.com',
7      pass: 'yourpassword'
8    }
9  });
10
11  var mailOptions = {
12    from: 'youremail@gmail.com',
13    to: 'myfriend@yahoo.com',
14    subject: 'Sending Email using Node.js',
15    text: 'That was easy!'
16  };
17
18  transporter.sendMail(mailOptions, function(error, info){
19    if (error) {
20      console.log(error);
21    } else {
22      console.log('Email sent: ' + info.response);
23    }
24  });
```



THANK YOU



BUILT-IN MODULE

IF YOU FIND THIS HELPFUL,
FOLLOW TO GET MORE.



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